Integrating societal preferences: Decentralization of the European energy network

Kristoffer Hedegaard Aden
Aarhus University
Student ID: 201905079

Abstract

Filler

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1. Introduction

In a world where the necessity of reducing the carbon emissions of our energy networks is no longer a question but a requirement careful and deliberate planning of the energy system of our future is needed. When the Paris Climate agreement was signed into action in 2015 it signaled the commitment of the United Nations to reducing their overall carbon footprint [1]. This common goal and understanding of the necessity of decarbonization was a crucial step towards a more sustainable world, and providing a common base understanding of what needs to be done, but also provides a massively complicated technological challenge that must be solved in order for the agreement to be fulfilled.

Somewhat unique to the issue of decarbonization in the European Union, is the fact that, although the Paris Climate Agreement signifies the common goal of decarbonization, The Union is still comprised of many different countries with many different political ideologies, laws, societal preferences and goals of their own. This means that although it has long been known that a shared and interconnected energy grid is the most efficient and economically preferable solution, as seen for example in [2], reality becomes a lot more complicated as the member nations also must have their own best interest in mind.

Historically the impact of the union being made up of self-centered actors has often been discarded when trying to construct the electricity grid of the future, if for no other reason than the fact that implementing social preferences and societal paradigms in strict mathematical models of the European energy system is a challenging problem [3]. These models that typically aim to reduce the overall cost of the system as the primary objective are often meant to be used used as the basis for developing the future energy network and legislation. However using this least-cost optimization approach without considering social paradigms in the European Union tends to produce system models that, although being economically efficient at achieving decarbonization targets, that are intrinsically tied to the assumption that the EU has complete control over the member nations individual energy grids.

This approach of reducing the optimization problems to purely techno-economic problems that prioritize cost reduction over feasibility has helped achieve significant insights into decarbonization strategies for the European Union [4], but tends to lead to the energy system becoming heavily centralized [5]. Centralization in the context of energy systems refer to the majority of resources being pooled into a few areas that are favorable to the system in some way, be that installing a lot of wind turbines in a region with a lot of wind, maintaining energy production in regions close to where the energy is needed or any number of other mechanisms that result in a proportionally large investment into infrastructure in a single region. While centralization as a strategy is often the most cost efficient strategy it leads to a suite of other challenges.

Two such challenges are those of physical security and and political feasibility. With the recent surge in so-called hybrid attacks on European energy infrastructure[6], the focus on physical security has increased. If a system is overly centralized it means that a large amount of critical infrastructure is installed on a relatively small area. This makes the infrastructure an obvious target for such hybrid attacks as a single hostile action can impact more infrastructure and more people. As for the political feasibility, nations who end up as net energy importers become less self sufficient and increasingly reliant on the net exporting regions. This shift in power-balance heavily favors the net exporters and as such may face resistance from net importing nations that wish to remain at a high degree of self sufficiency.

The objective of this masters thesis is to incorporate societal paradigms and priorities in the form of decentralization into a macro-scale energy network model of the EU, and to analyze and expose the mechanisms behind decarbonization and its interactions with decentralization. The main hypothesis is that, by incorporating decentralization, the resulting networks will illuminate alternative transition pathways. This in tun may then help develop a workflow that by incorporating societal paradigms and reviewing their system impacts, will help close the gap between the models used to inform energy policy makers and the results of the proposed policies. Although this workflow and methodology may in time prove relevant for policy makers to understand, the target audience of this master's thesis remains peers in the field of energy system modeling who may draw inspiration from the intersection between societal paradigms and strict mathematic modeling that this work represents. It is also for this reason that all the data and code used to produce the results which can be seen in this paper is openly available in a public repository under an open science license[7].

2. Nomenclature

Symbols

Symbol	Description
n	Node index (spatial unit)
t	Time snapshot index
g	Generator technology index
C	Decarbonization level (fraction of baseline emissions)
K	Decentralization parameter (capacity distribution factor)
$\lambda_{n,t}$	Marginal electricity price at node n and time t (EUR/MWh)
μ_C	Shadow price of CO_2 emissions at decarbonization level C (EUR/tCO_2)
p	Power output or capacity (MW)
e_g	Emission factor for generator technology g (tCO_2/MWh)
w_t	Temporal weighting factor for snapshot t
$L_{n,t}$	Electrical load (demand) at node n and time t (MWh)
ω_n	Normalized load-share weight for node n
R	Renewable generation capacity (MW)
r	Discount rate (per annum)
G	Gini coefficient of inequality

Abbreviations

Abbreviation	Definition
AC	Alternating Current
CAPEX	Capital Expenditure
CCGT	Combined Cycle Gas Turbine
EU	European Union
FOM	Fixed Operation and Maintenance
HPC	High Performance Computing
HVDC	High Voltage Direct Current
KKT	Karush-Kuhn-Tucker
LCOE	Levelized Cost of Electricity
OCGT	Open Cycle Gas Turbine
PHS	Pumped Hydro Storage
PyPSA	Python for Power System Analysis
SDRCL	Standard Deviation of Nodal Renewable Capacity per Load

3. Methodology

3.1. Modeling framework

For the purpose of structuring the modeling a Snakemake workflow was constructed. Snakemake is a workflow management system that allows for modular and reproducible data analysis [8]. The workflow is divided into multiple stps or "rules" that when chained together ensures that the data is processed in the correct order. A workflow management system was implemented for two reasons. As docmented later on in the thesis the large number of scenarios simulated with the workflow lends itself well to a standardized workflow that can handle small variations two construict different scenarios. Secondarily Snakemake is already used in many power system optimmization workflows and was therefore the obvious candidate. The complete workflow used to produce the results in this master's thesis is openly available in a public repository [9] under a GNU General Public License [10].

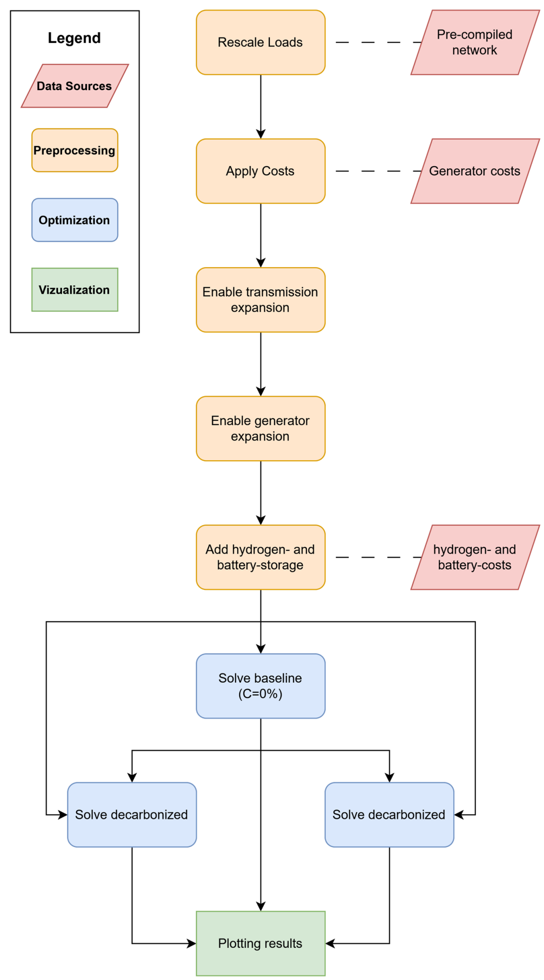

Figure 1: Map of Snakemake Workflow

Seen above in Figure 1 is a map of the rules and data-input used in the workflow divided into three main sections: Preprocessing, optimization and vizualization/postprocessing. The following sections will go further into detail on the diferent sections of the workflow, excluding the vizualizations and postprocessing which are primarily used for producing figures and tables for this master's thesis.

3.2. Precompiled network model

Many energy modeling frameworks and pieces of commercially available software for optimizing nodal-networks exist. The work represented by this paper utilizes the open-source framework PyPSA (Python for power system analysis [11]). The reasoning for this is twofold: Firstly PyPSA is completely free and open source to use. This encourages open and collaborative science as well as transparency in the philosophy behind the energy system modeling. Secondly PyPSA consists of two separate parts. A modeling workflow based in the Python programming language, and a large database containing relevant open source data for constructing models, particularly of the European Union. This dataset is referred to as PyPSA-Eur[12] and is invaluable in its quality of data for building macro energy system models in the PyPSA workflow. This masters thesis utilizes both parts of the PyPSA environment. Acting as a baseline for all of the network topology for the European Union is a pre-compiled network developed with PyPSA-EUR using the default settings as published in the Zenodo repository [13].

The simple philosophy behind the PyPSA models is that a network is perceived as multiple interconnected nodes. These nodes and connections are then given a set of constraints that define the feasible solution space. Examples of constraints could be that each node is assigned an electrical load that must be serviced based on historical consumption, capacity factor time-series based on solar- and wind-patterns, or maximum or minimum capacity constraints like a maximum allowable power flow from one node to another representing the transmission infrastructure in the network. The network used in this master's thesis is a 37 node network representing the 25 (excluding the island states of Malta and Cyprus) member nations of the EU. Additionally Switzerland, Norway, Bosnia-Herzegovina, Albania, Montenegro, North Macedonia, The United Kingdom and Serbia are included as their electrical infrastructure is connected to that of the EU. The following nations are also represented by more than one node as they have large populations on islands: Denmark, Spain, Great Britain and Italy.

These nodes are then interconnected with HVDC and AC links and lines based on real-world connections. A graphical overview of the network can be seen below in Figure 3.2:

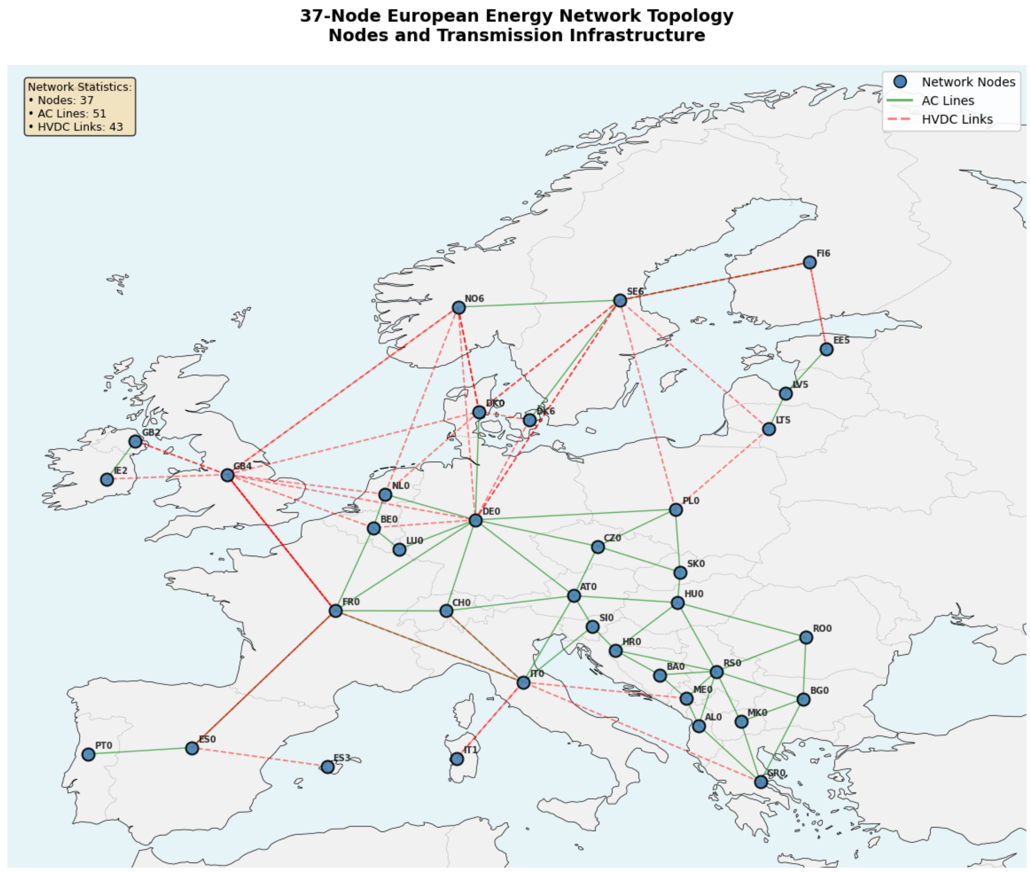

Figure 2: Overview of the 37-node European energy network.

The 37- node network is in actuality a clustered version of the standard PyPSA-EUR network which contains 1024, which is incredibly computationally intensive to solve. The effects of the clustering process on the network have been analyzed in [14], where it is found that similar power system models show relatively small sensitivity to increases and decreases in the number of nodes.

3.3. Load rescaling

The precompiled network models comes preloaded with electrical demand time series for each of the nodes from the ENTSOE-E transparency platform

[15]. These hourly demand time series are, however, in the case of the pre-compiled model from the year 2013. In order to ensure a more representative demand profile for the nodes the loads have been rescaled to projected values for 2025. Each node is assigned a target electricity consumption value based on figures from Carbon-Free Europe [16]. These are then used together with the load values from 2013 to calculate a nodal scaling factor for each node as seen in equation (1) below:

$$\text{scaling factor}_n = \frac{\text{target Load}_n}{\text{2013 Load}_n} \quad \forall\, n \tag{1}$$

Here n denotes the set of all nodes. These nodal scaling factors are then used as weights on the hourly load time series for each node to produce a rescaled load time series for each node as seen in equation (2) below:

$$\text{rescaled Load}_{n,t} = \text{scaling factor}_n \cdot \text{2013 Load}_{n,t} \quad \forall\, n, t \tag{2}$$

Where t denotes the set of all time snapshots. This way the total load is scaled to the projected value but the general shape of the load time series is preserved. An example of a nodal load time series before and after rescaling can be seen below in Figure 3:

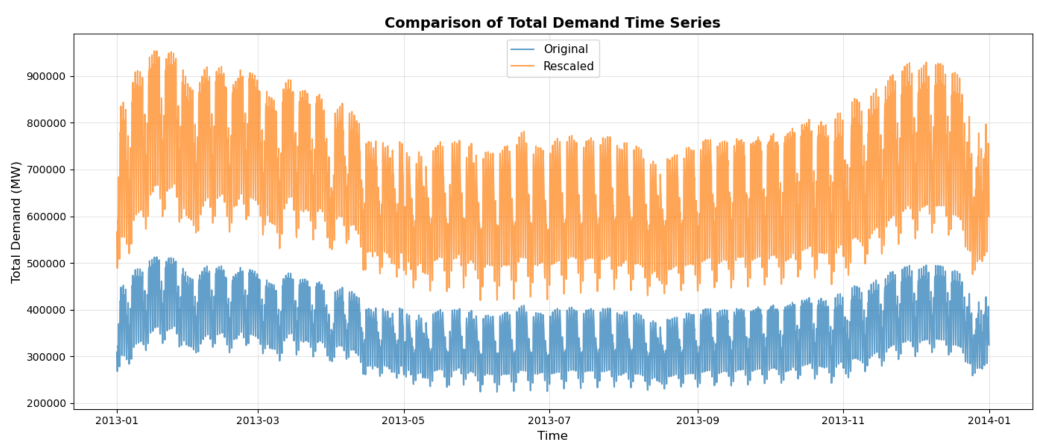

Figure 3: Example of load rescaling for node representing Denmark.

Here it can be seen that the shape of the load time series has been preserved while the total load has been rescaled to match the projected values.

This rescaling process assumes that while the total energy consu,ption will increase the pattern will remain constant. This is a simplification as continuing electrification of different sectors such as heating and transport, as well as changing weather patterns due to climate change, may very well alter the consumption pattern in the future. However, sector coupling and synthetic climate change models aare beyond the scope of this master's thesis and this simplification is therefore necessary.

3.4. Generator Cost Modeling

Although the precompiled network comes with some cost data already defined, there are some technologies that do not hace cost data defined. In order to be able to optimize the network fully this must be rectified. Table 1 below shows the cost data used for the different generation and storage technologies in the network.

Table 1: Cost assumptions for generation technologies [17] [18] [19]

Carrier	**CAPEX (EUR/MW)**	**Lifetime (years)**	**Discount rate**	**FOM (abs) (EUR/MW/a)**	**FOM (rate) (per a)**	**Notes**
CCGT	830,000	25	0.07	27,800	–	DEA 2030
OCGT	435,000	25	0.07	7,745	–	DEA 2030
Coal	1,860,000	30	0.07	30,355	–	DEA 2030
Lignite	1,860,000	30	0.07	30,355	–	Proxied as coal CAPEX
Oil	343,000	25	0.07	8,448	–	DEA 2030
Biomass	3,300,000	25	0.07	96,000	–	DEA 2030
Geothermal	4,290,000	20	0.07	–	0.02	EGS ∼2% FOM
Nuclear	5,870,000	60	0.07	–	0.02	NEA 2020 ∼2% FOM

PyPSA, stores generator cost paramaters as annualized total ownership costs. Due to differences in the way different sources report costs, some conversion is necessary in order to convert the reported costs into the format used by PyPSA. Firstly, for sources that report the FOM (Fixed Operation and Maintenance) costs as an absolute value the total annualized cost is calculated as:

$$\text{Annualized Cost} = \text{CAPEX} \cdot \text{Annuity Factor} + \text{FOM (abs)} \qquad (3)$$

Where the Annuity Factor is calculated as:

$$\text{Annuity Factor} = \frac{r(1+r)^n}{(1+r)^n - 1} \tag{4}$$

With r being the discount rate and n being the lifetime of the technology in years. For sources that report FOM as an annual rate the annualized cost is calculated as:

$$\text{Annualized Cost} = \text{CAPEX} \cdot \text{Annuity Factor} + \text{CAPEX} \cdot \text{FOM (rate)} \tag{5}$$

An finally for sources that report neither FOM (abs) nor FOM (rate) the annualized cost is simply calculated as calculated as:

$$\text{Annualized Cost} = \text{CAPEX} \cdot \text{Annuity Factor} \tag{6}$$

Performing these operations ensures that all the cost data is formatted the same when interfacing with the PyPSA model. The full cost assumptions for the different types of generators can be found in Appendix A: Generator cost assumptions. It os also important to note that anuallization is in fact a specific case of scaling costs over tme, and the yearly cost is simply because the optimization timespan inclded in the pre-compiled model is a single year.

3.5. Transmission- and generator-expansion

In order to find the optimal network the transmission infrastructure as well as select generator types must be allowed to vary their capacity. For transmission infrastructure the line capacity was constrained as seen in equation (7) below:

$$p_{\text{existing}} \leq p_{\text{line}} \leq p_{\text{line, max}} \tag{7}$$

Where p_{existing} is the existing capacity of the line, p_{line} is the optimized capacity of the line and $p_{\text{line, max}}$ is the maximum allowable capacity of the line. The minimum capacity is set to the pre-existing capacity as it is assumed that the existing transmission infrastructure will not be decommissioneddue to the trouble of digging up existing lines. In order to avoid transmission bottlenecks, and find the optimal solution of installed generation capacity in the nodes, the maximum capacity was set to an arbitrarily high value.

A similar approach was used for the *fossil-based* generators in the system. These generators are allowed to decomission in order to meet the decarbonization targets, but are not allowed to expand their capacity beyond the existing existing capacity as can be seen in equation (8) below:

$$p_{\text{nom}} \leq p_{\text{existing}} \tag{8}$$

Most of the renewable generators are allowed to expand their capacity between 0 and an arbitrarily high value as these are the capacities that will later be constrained by the decentralization measures. However some renewable generators like nuclear, geothermal and biomass are, constrained to "only" expand a factor 100 beyond their existing capacity int the nodes where they are already present. This is primarily done as calculating the maximum installable capacity for these technologies is a large task in itself and is outside the scope of this master's thesis. The freely expandable renewable technologies in the system are therefore solar PV, onshore wind and offshore wind which are free to expand and contract with every optimization, as long as the capacity is positive as seen in equation (9) below:

$$0 \leq p_{\text{nom}} \tag{9}$$

In addition to the capacity constraints, the operation of the different generators is controlled by availability factors. Each generator's maximum power output at any at any time is given as ($p_{\text{max\_pu},g,n,t}$) and the minimum stable generation ($p_{\text{min\_pu},g,n,t}$).

3.6. Energy storage infrastructure

Finally, the last step before the system is ready to be optimized, is to implement the energy storage infrastructure. The pre-compiled network comes with a few different types of storage already implemented: Pumped-Hydro Storage (PHS), and Regular (resovoir) hydro storage. Historically hydro based storage in the European Union has been the primary method of energy storage and as a result most areas capable having hydro based storage installed have already been utilized. As a result of this the hydro based storage technologies are not allowed to change their capacity in the optimization and are constrained as seen in equation (10) below:

$$p_{\text{nom}} = p_{\text{existing}} \tag{10}$$

However as the system decarbonizes and the amount of dispatchable fossil-based generation decreases it is increasingly necessary to have large amounts of energy storage capacity installed. In order to properly model this two additional storage technologies, namely Battery storage and hydrogen storage has been added to the network. Battery stoage consists of battteries and inverters and hydrogen storage consists of a hydrogen tank, an electrolyzer as well as a fuel cell. When adding the additional storage technologies their costs have been calculated in a manner similar to that o the generators as seen in section 3.4 above. The cost assumptions for battery and hydrogen storage can be seen below in Table 2:

Table 2: Storage technology cost assumptions used in the optimization model [20].

Component	**Type**	**CAPEX (EUR/unit)**	**Lifetime (years)**	**Efficiency**	**Annualized Cost (EUR/unit/a)**
Battery	Energy (Cells)	142 kWh	25	—	12.2k MWh/yr
	Power (Inverter)	160 kW	25	0.96	13.7k MW/yr
Hydrogen	Energy (Tank)	57 kWh	25	—	4.9k MWh/yr
	Power (Electrolyser)	600 kW	30	0.66	48.4k MW/yr
	Power (Fuel Cell)	1100 kW	10	0.50	156.6k MW/yr

Here it is important to note that the storage units include effciencies for charging and dischaging. This efficiency is incoporated in optimization model as it has a large imnpact on system cost, which was annualized as per equations in section 3.4 above. As for the complete cost assumptions for the different storage components these can be found in Appendix B: Storage cost assumptions.

3.7. Optimization strategy

Optimizing complex energy networks with many variables and constraints is computationally very intensive. For this reason the optimizations performed for this master's thesis have exclusively been performed on a High Performance Computing (HPC) environment. More specifically the UCloud HPC operated by SDU [21]. The optimizations were typically run on a set of 4 machines, each with an Intel Xeon Gold 6130 CPU and took on average 12 core-hours apiece to achieve an optimal solution. The optimization algorithm was chosen as the Barrier method in the commercial optimizer by Gurobi, chosen primarily for its numerical stability [22].

3.8. Baseline system

As seen in Figure 1 above, in order to be able to analyze the impacts of varying decarbonization constraints and decentralization measures first a baseline system that is subjected to neither decarbonization nor decentralization constraints must be optimized. This system, which will be used for comparrison is referred to as the *baseline* system. The optimization of this system is performed based on the objective function and load-balance, and is a *least cost* system, meaning that the total system cost is minimized.

In order to optimize the networks first an objective function must be defined. An objective function is, as the name suggests, the function which when minimized or maximized leads to the objective of the optimization to be achieved. The objective function for finding the cheapest possible electrical system is defined as minimizing the sum of all the costs:

Equation:

$$\min\Big(\sum_{n} \text{Capital costs} + \sum_{n,t} \text{variable costs}\Big) \tag{11}$$

Here n denotes the set of all nodes, and t denotes the set of all snapshots during the timeframe. The capital costs reflect things such as the cost of building generation, storage and transmission capacity and the variable costs reflect things such as fuel costs and maintenance. In order to construct a feasible solution space the objective function must be defined by a set of bounding constraints. Initially an equality constraint is defined so that, for each snapshot, all nodes must have their electricity demand met by a combination of energy production and nodal balancing:

$$\text{generation}_{n,t} + \text{balance}_{n,t} = \text{demand}_{n,t} \quad \longleftrightarrow \quad \lambda_{n,t} \quad \forall\, n,t \tag{12}$$

Here λ is known as the dual-variable or the Karush-Kuhn-Tucker (KKT) multipler associated wth the constraint. This variable can be interpreted as the shadow price, or marginal price of electricity in node n at time t. This marginal price represents the cost of supplying one additional unit of electricity to a node and as such is set by the most expensive generator needed to meet demand in that node at that time[23].

Finding the dual variable is done by formulating a Lagrangian function that relates the gradiant of the objective function to the gradients of the constraint:

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{n,t} \lambda_{n,t} g_{n,t}(x) \tag{13}$$

Here $f(x)$ is the objective function, $g_{n,t}(x)$ is the constraint function and x is the set of all optimization variables.The differnt KKT multipliers can then be found by optimizing the Lagrangian function with respect to both the optimization variables and the KKT multipliers and thus finding the point where the gradients are zero as seen in equation (14) below:

$$\nabla_x \mathcal{L}(x^*, \lambda^*) = 0 \quad \text{and} \quad \nabla_\lambda \mathcal{L}(x^*, \lambda^*) = 0 \tag{14}$$

Here the asterisk denotes the optimal values of the optimization variables and the KKT multipliers respectively.Optimizing the network without a decarbonisatio constraint, is then used to produce a baseline system that will be used for comparrison in the following sections.

3.9. Decarbonized systems

Once the baseline system has been optimized the a set of new networks at different levels of decarbonization can be defined and optimzed. In addition to the objective function (11) and the load balance (12) these systems are also constrained by a decarbonization constraint as seen in equation (15) below:

$$\text{total emissions}_C \leq (1 - C \cdot \text{baseline emissions}) \quad \longleftrightarrow \quad \mu_C \tag{15}$$

Where C is the level of decarbonization as a fraction of the baseline emissions (the total emissions from the baseline system) and μ is the KKT multiplier and can be interpreted as the price of emitting an additional unit of $\mathrm{CO_2}$ in the system at decarbonization level C. This is aditionally equal to setting the CO_2 price equal to the dual variable and optimizing. The baseline emissions are calculated based on the emission factors of each generator technology and their respective power output as seen in equation (16) below:

$$\text{total emissions} = \sum_t \sum_n \left(\sum_g p_{g,n,t} \cdot e_g \right) \cdot w_t \tag{16}$$

Here p is the power output of generator g in node n at time t, e is the emission factor of generator g and w is the weighting factor for snapshot t.

The emission factors for the different technologies, are pre-loaded in PyPSA and can be seen in table 3 below:

Table 3: Emission factors by carrier technology

Carrier	**Emission Factor (tCO_2/MWh_{fuel})**
CCGT	0.1870
OCGT	0.1870
coal	0.3540
geothermal	0.0260
lignite	0.3340
oil	0.2480

Based on previous experience with opimising energy systems the following levels of C were chosen for the reference systems:

$$C \in \{0\%, 10\%, 20\%, 30\%, 40\%, 50\%, 60\%, 70\%, 80\%, 90\%, 93\%, 96\%, 99\%, 100\%\} \tag{17}$$

This was done in order to achive a wide range of decarbonization, while increasing resolution at higher levels of decarbonisation where the system tends to change more rapidly. The outcome of the the decarbonization workflow is therefore a set of 14 different networks (excluding the baseline), each at a different level of decarbonization defined as a fraction of the baseline emissions.

3.10. Decentralized systems

As discussed in the introduction, decentralization is necesarry in order to ensure a fair and secure european energy system. The first step in creating a set of decentralized systems, at each level of decarbonization, is to define a measure of decentralization. Some papers like "Modeling all alternative solutions for highly renewable energy systems" by Pedersen et al. [24] use the *Gini Coefficient* of inequality of installed renewable capacity as a measure of decentralization. Consider the following plot:

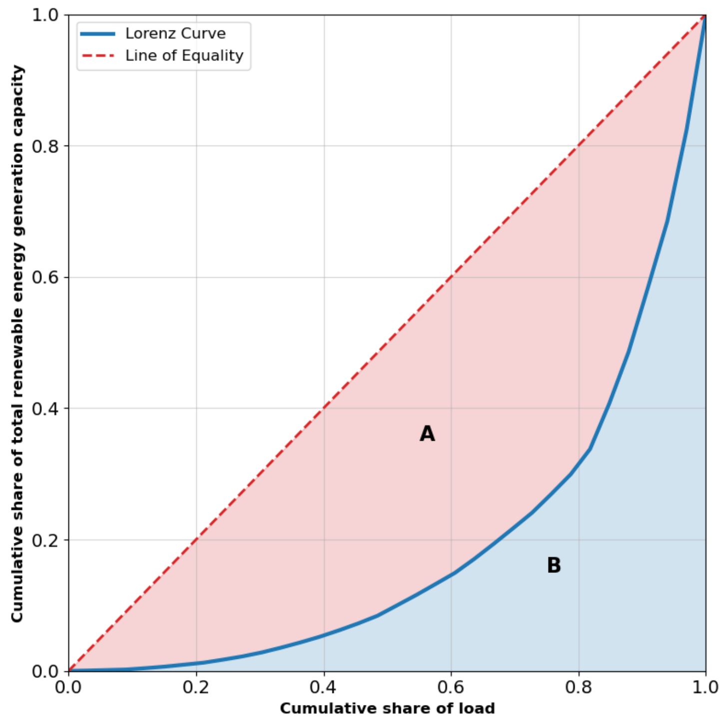

Figure 4: Example of Gini coefficient calculation for installed renewable capacity.

Here the red line represents an equal distribution of renewable capacity where the cumullative share of renewable energy generation capacity is equal to the cumullative share of the load for the same nodes. The blue line is an example of a Lorentz-curve where some nodes have a higher share of the renewable capacity than their share of the load (further to the right on the x-axis) while other nodes have a lower share of the renewable capacity than their share of the load (further to the left on the x-axis). The total inequality in the system can then be quantified by calculating the Gini coefficient as seen in equation (18) below:

$$G = \frac{A}{A + B} \tag{18}$$

Where A is the area-difference between the equality line and the Lorentz curve, and B is the area-differnce between the Lorentz curve and the x-axis. This way of defining system inequality comes with a number of issues.

Primary among these issues is that fact that the Gini coefficient is non-linear and as such cannot directly be used as a constraint as this will lead to non-convex solution spaces and local minima. While it is possible to work around this by using a sampling based approach and then calculating the Gini coefficient post optimization, this approach is difficult to understand and implement. As a result the decentralization of the systems was constrained as seen in equation (19) below:

$$\frac{1}{K} \cdot \omega_n \cdot R_{\text{total}} \leq R_{\text{cap},n} \leq K \cdot \omega_n \cdot R_{\text{total}} \quad \forall\, n \tag{19}$$

Where $\omega_n = \frac{L_n}{\sum_{n'} L_{n'}}$ is the normalized load-share weight for node n, R_{total} is the total renewable capacity across all nodes, $R_{\text{cap},n}$ is the installed renewable capacity at node n, and K is the decentralization parameter. This way each node is given a minimum renewable capacity as a fraction of the total renewable capacity in the system based on their share of the total load. Simultaniously each node is also assigned a maximum renewable capacity as a factor of the total renewable capacity in the system based on their share of the total load. If, for examle $K = 3$ that means that each node is allowed to "outperform" their load-share based renewable capacity by a factor of 3, while also being required to install at least one third of their load-share based renewable capacity. By then gradually having K approach 1 the system is foced to decentralized until at $K = 1$ each node is required to install renewable capacity exactly equal to their load-share based renewable capacity, which leads to the system becoming increasingly decentralized.

Based on previous preliminary experiments following levels of K were chosen for the decentralized systems:

$$K \in \{8, 7, 6, 5, 4, 3, 2, 1\} \tag{20}$$

This approach has the benifit of being linear, but also avoids another pitfall when implementing decenralization in systems that also are subjected to decarbonization constraints. Both these constraints, in some way, interact with the amount of renewable capacity installed in the nodes of the network. If one was to simply constrain the amount of installed renewable capacity in each node without taking into account the total amount of renewable capacity in the system, it could lead to the system installing more renewable capacity than needed to meet the decarbonization constraint. This interaction would then lead to the system becoming overly decarbonized and difficult to compare to other systems.

The final outcome of the decentralization workflow is therefore a set of 105 different networks, each at a different level of decarbonization and decentralization. Before proceeding to the results of the optimization workflow it is usefull to introduce a more intuitive measure of the decentralization of a given system as the K parameter is somewhat abstract. A more intuitive measure of decentralization is the *Standard Deviation of Nodal Renewable Capacity per Load* defined as:

$$\text{SDRCL} = \sqrt{\frac{1}{n}\sum_{n}\left(\frac{R_{\text{cap},n}}{L_n} - \mu\right)^2} \quad \text{where} \quad \mu = \frac{1}{n}\sum_{n}\frac{R_{\text{cap},n}}{L_n} \tag{21}$$

Here $R_{\text{cap},n}$ is the installed renewable capacity at node n, L_n is the load at node n and n is the number of nodes. This measure represents the standard deviation of the ratio between installed renewable capacity and load at each node. A low value of SDRCL indicates that the renewable capacity is more evenly distributed across the nodes relative to their load, while a high value indicates a more uneven distribution. By then comparing the SDRCL values for each of the decentralized systems to the network at the same level of decarbonization but without decentralization constraints, it is possible to get a more intuitive measure of how decentralized a given system is and how succesfull the decentralization constraints were.

4. Results

In the following section the results from the workflow discussed in the methodology serction will be shown and analyzed. Starting the the baseline system, followed by the decarbonized systems and finally the decentralized systems. After each type of system the underlying mechanisms and interactions present in the systems will be discussed.

4.1. Baseline system

As a reminder the baseline system is purely constrained by the objective function (11) and the load balance constraint (12). Figure 5 below shows a map of the installed generation capacities for the baseline system:

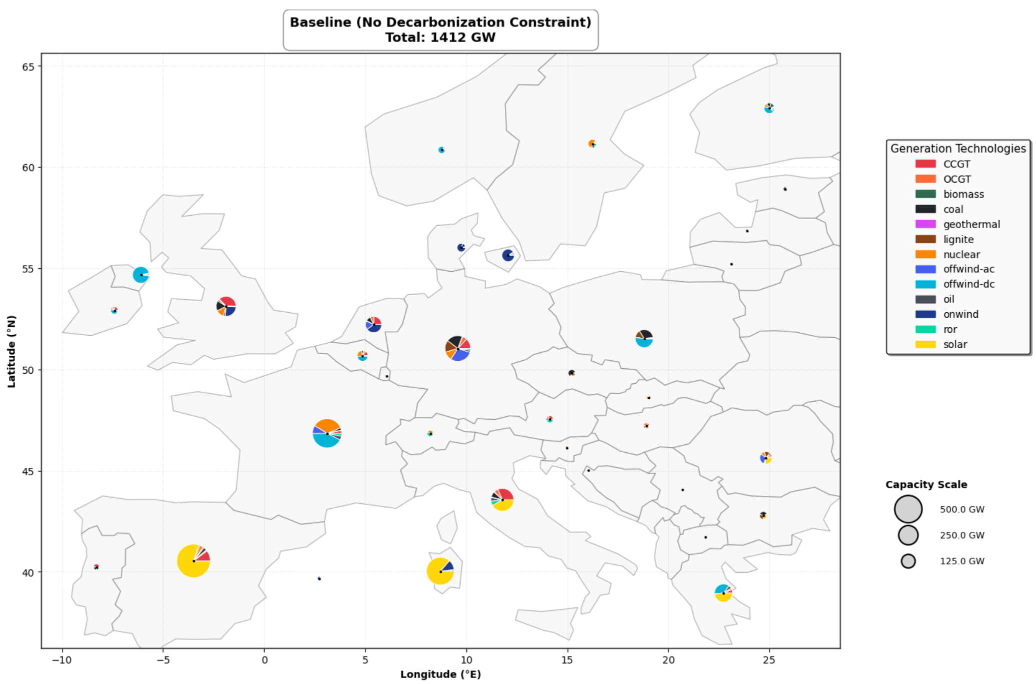

Figure 5: Baseline generator capacity map.

Here it can be seen that the gap between the load and the fossil generation has been filled with renewable generation as the fossil based generation technologies are not allowed to expand their capacity. The total installed generation cpacities in the baseline system can be seen in figure 6 below:

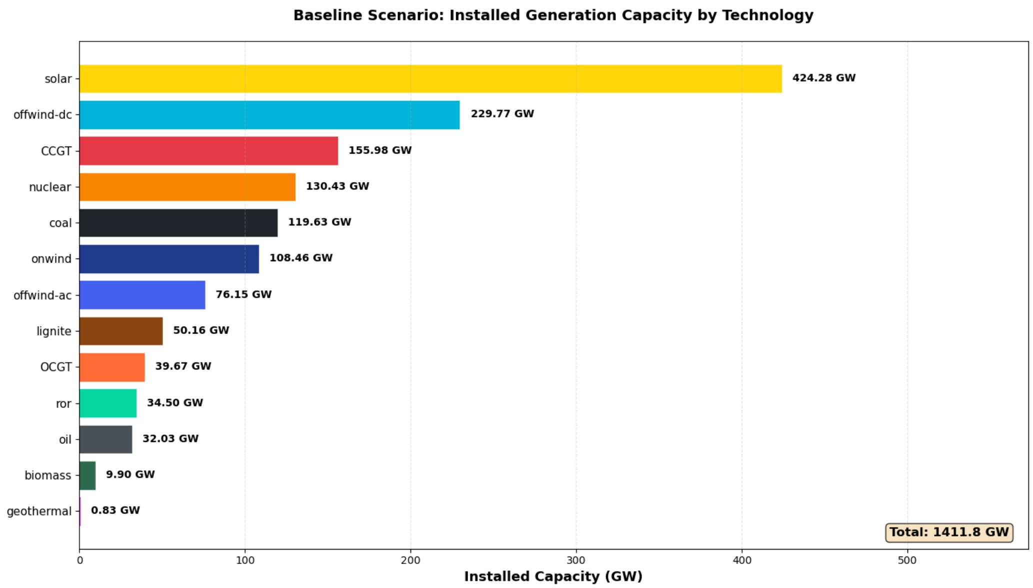

Figure 6: Baseline generator capacity by technology.

It is however important not to conflate the installed capacities with the actual generation as technologies differ in capacity factor time series and utilization. The actual generation by technology can be seen below in figure 7:

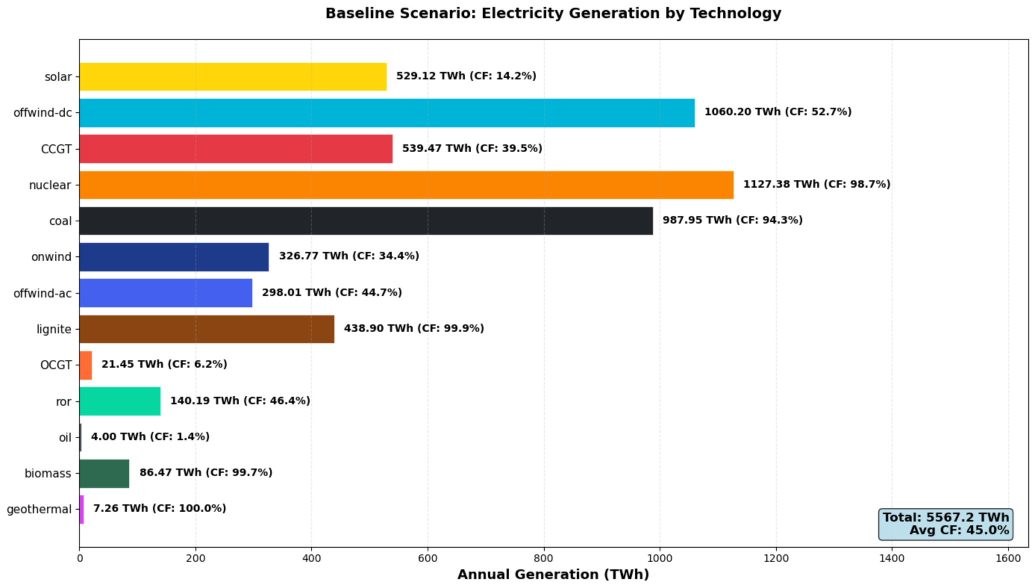

Figure 7: Baseline generator output by technology.

As for the transmission capacities these can be seen for the baseline scenario in figure 8 below:

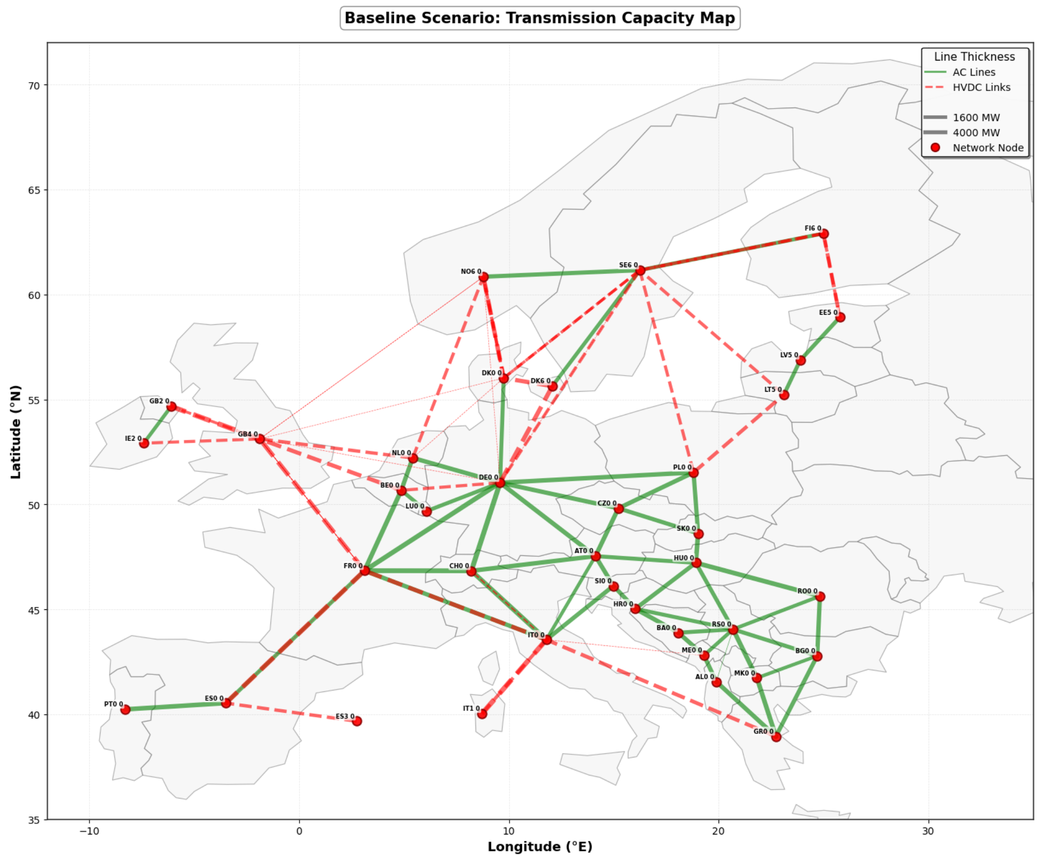

Figure 8: Baseline transmission capacity map.

The transmission infrastructure in the baseline system conntains 490.2 GW HVDC capacity and 232.4 GW AC capacity. Since a large amount of the load is now serviced by renewable generation, whose capacity factor time series do not necessarily line up with the load time series, storage infrastructure is used to de-couple supply and demand temporally. The installed storage capacities totalling 650 GW in the baseline system can be seen below in figure 9:

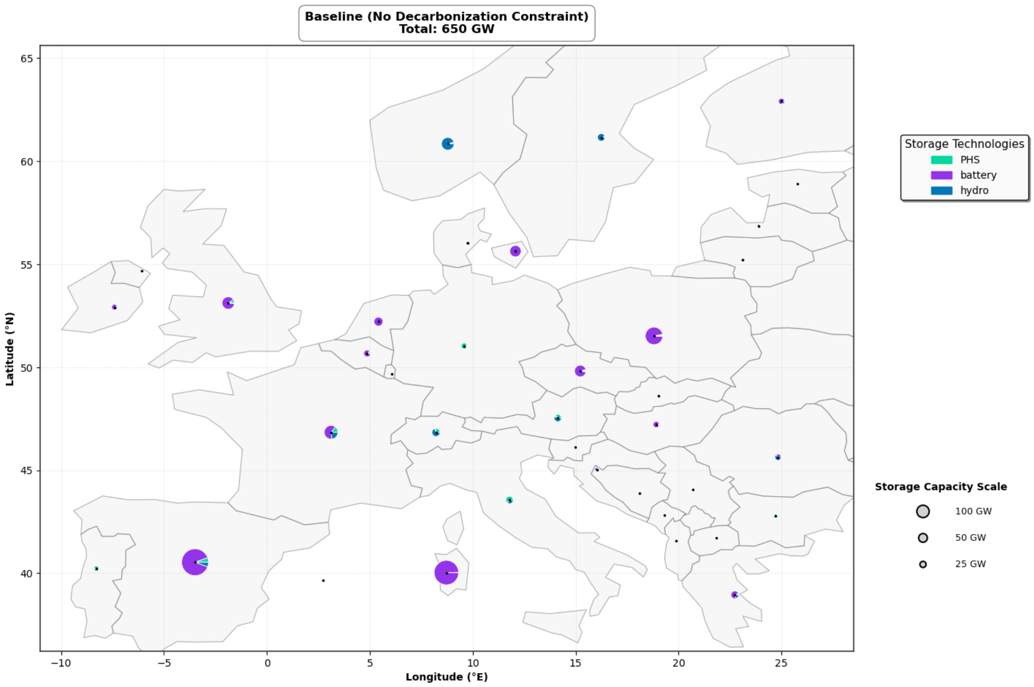

Figure 9: Baseline storage capacity map.

Here it can be seen that the optimizer resulted in a very substantial buildout of battery storage capacity while the hydrogen storage capacity is absent. This will of course be addressed in more detail in the analysis section below. The total emissions from the baseline system as calculated by equation (16) is 1297 $MtCO_2$. The breakdown of which technologies contribute to these emissions can be seen below in figure 10:

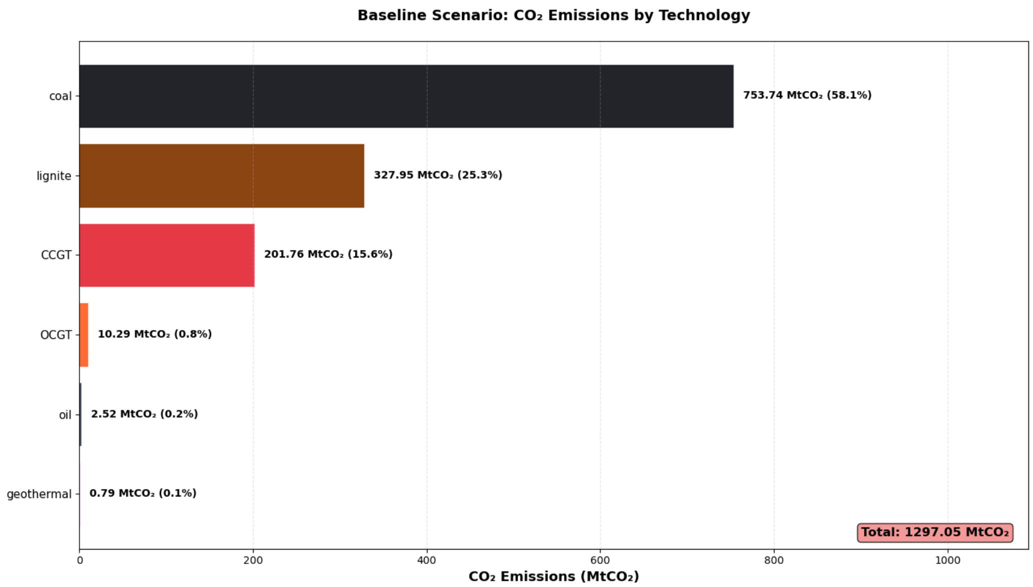

Figure 10: Baseline emissions by technology.

Here the two solid-fuels coal and lignite are the major contributors to the emissions. This is to be expected when considering their high emission factors as seen in table 3 and their large generation contributions as seen in figure 7 above.

For the economic side of the baseline system the mean marginal demand-weighted price and the total system cost are of interest. These metrics can be seen below in table 4:

Table 4: Baseline System Economic Metrics

Metric	**Value**
Mean Marginal Price (Demand-Weighted)	68.56 €/MWh
Total System Cost	348.18 B€/year

Here The mean marginal demand-weighted price is defined as seen in equation (22) below:

$$\text{Mean Marginal Demand-Weighted Price} = \frac{\sum_{t \in T} \sum_{n} (L_{n,t} \times \lambda_t)}{\sum_{t \in T} \sum_{n} L_{n,t}} \tag{22}$$

For time t, Nodes n, Loads L and marginal price λ. This cost measure represents the average price that consumers pay for electricity over the entire time period, weighted by their demand. The total system cost is simply the sum of all costs as minimized by the objective function (11).

Finally, the system is, visually, as seen in figure 5 above quite centralized. This can be quantified by calculating the Standard Deviation of Nodal Renewable Capacity per Load (SDRCL) as defined in equation (21). For the baseline system the SDRCL is calculated to be 11.84 $\mathrm{MW}_{\mathrm{renewable}}/\mathrm{MW}_{\mathrm{load}}$. This figure is difficult to quantify in isolation as its intended use is as a comparrision metric. However by vuzalizing the nodal renewable capacity per load in figure 11 below it is possible to get an idea of how for centralized ohe baseline system is.

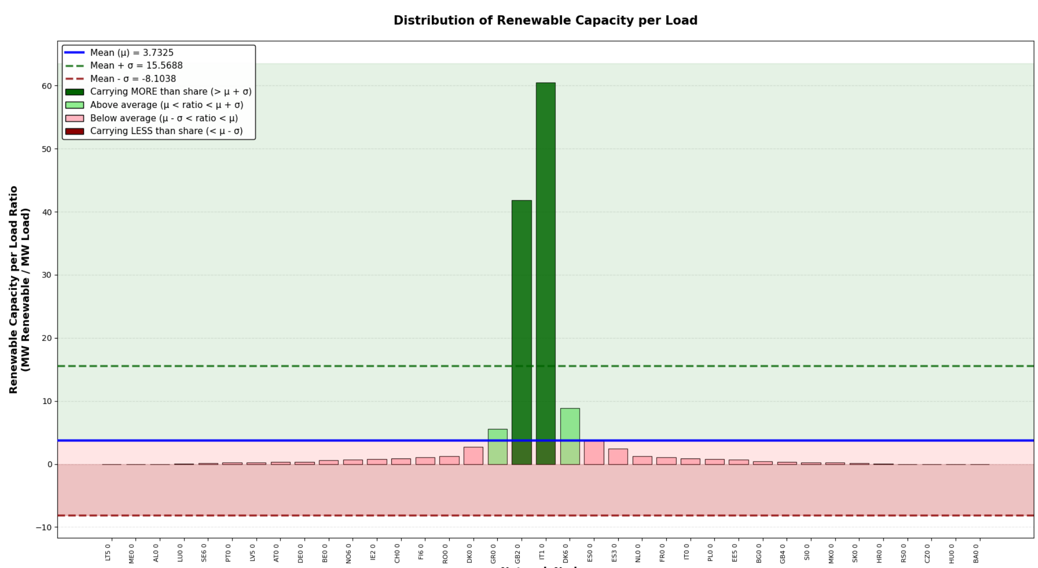

Figure 11: Baseline renewable capacity per load map.

Here it can be seen that only four nodes (IT0, GB2, GR0 and DK6) have a larger installed renewable capacity than their yearly load. The relevant statistics for the centralization of the baseline system can be seen in table 5 below:

Table 5: Baseline System: Renewable Capacity per Load Distribution Statistics

Metric	**Value**
Total Nodes (with load)	37
Mean Ratio ($MW_{renewable}/MW_{load}$)	3.73
Standard Deviation (SDRCL, σ)	11.84
Upper Bound ($\mu + \sigma$)	15.57
Lower Bound ($\mu - \sigma$)	-8.10
Distribution:	
Nodes Above $\mu + \sigma$	2
Nodes Within $\pm\sigma$	35
Nodes Below $\mu - \sigma$	0

4.2. Baseline mechanisms and interactions

The baseline system behaviour is, as can be seen above, fairly intuitive. As the pre-compiled network is a *Brownfield* system that contains existing fossil-based generation capacity but no renewable based generation capacity, the optimizer services the load with the renewables as the fossil based generators are not allowed to expand their capacity. The large capacity builout is domnated by solar PV in the south of Europe and off-shore and on-shore wind in the northern regions. As for the actual generation, as can be seen in figure 7 above, the massive solar buildout has not converted into a similarly large generation contrbution. This is primarily caused by the fact that solar PV is more limited by its capacity factor time series than wind based generation. The largets contributor to the generation ends up becoming nuclear power, which has a high capacity factor time series and is not limited by weather conditions. In addition nuclear, is dispatchable and therefor does not require storage to the same extent as weather dependent generation.

Transmission infrastructure has also been expanded in order to transport the energy from non-dispatchable renewables to the load centers where it is needed. As for storage infrastructure the optimizer has all but neglected to install hydrogen based storace capacity while expanding battery storage massively. This is primarily caused by the high round trip efficiency loss as seen in table 2 above. This large loss leeds to hydrogen based storage becoming prohibitively expensive compared to battery storage. The lack of hydrogen based storage also means that the system is less likely to utilize wind-based renewable generation compared to solar based generation as wind generation tends to be more seasonal better served by hydrogen storage.

The baseline emissions, are as explained above, dominated by solid fuel based fossil generation technologies due to their high emission factors. The mean marginal demand-weighted price of electricity in the baseline system is calculated to be 68.56 €/MWh which is fairly reasonabl and well within the same order of magnitude as found in papers on similar models such as in [5].

Finally the baseline system as seen both visually in figure 11 perfectly encapsulates the need for decentralization measures. While the heavily centralized nature of the system is optimal from a simple least-cost perspective, it is likely to lead to issues with public acceptance, political feasibility and security.

4.3. Decarbonized systems

Since the outcome of the decarbonization worklow is the baseline plus 14 systems , each for a different level of decarbonization, it is more useful to vizualize the evolution for the key metrics, rather than go through them one by one. Seen below is the mix of installed generation capacity for the different levels of decarbonization in figure 12:

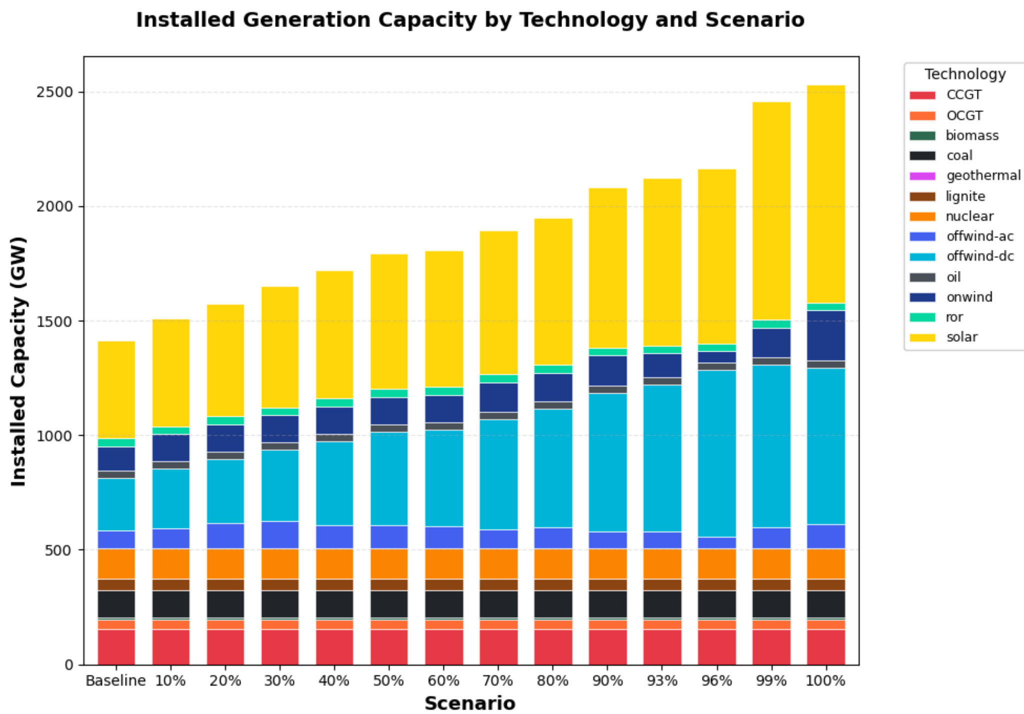

Figure 12: Decarbonized generator capacity by technology.

Here it can be seen that in order to decarbonise the system completely the total installed generation capacity increases from around 1400 GW in the baseline system to around 2500 GW in the fully decarbonized system, or a around a 78% increase. As expected the fossil based generation capacity does not dissapear as the system is not forced to decomission aexisting capacity, but rather to replace the actual generation from fossil based generators with renewable generation. The actual generation breakdown for the different levels of decarbonization can be seen below in figure 13:

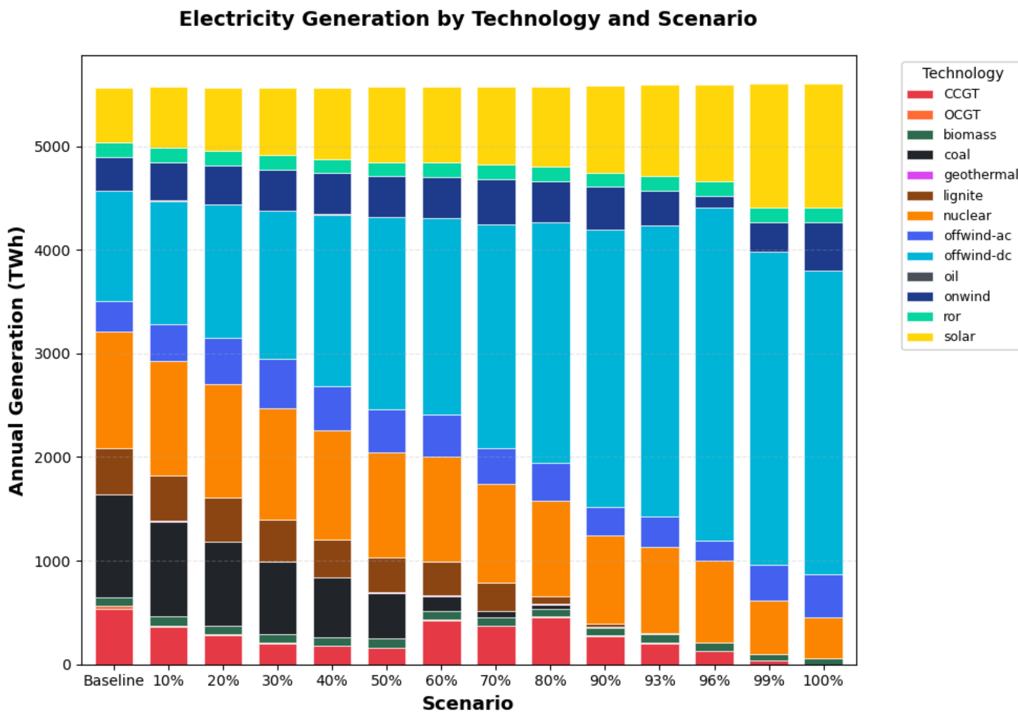

Figure 13: Decarbonized generator output by technology.

Here it can be seeb that as the system decarbonizes the fossil fuel based elecetricity generation goes to zero and is replaced primarily with offshore wind and solar PV generation. In order to better vizualize the evolution of the generation of the diffrent technolgies as the system decarbonizes, seen below in figure 14 is a line plot of the generation by technology:

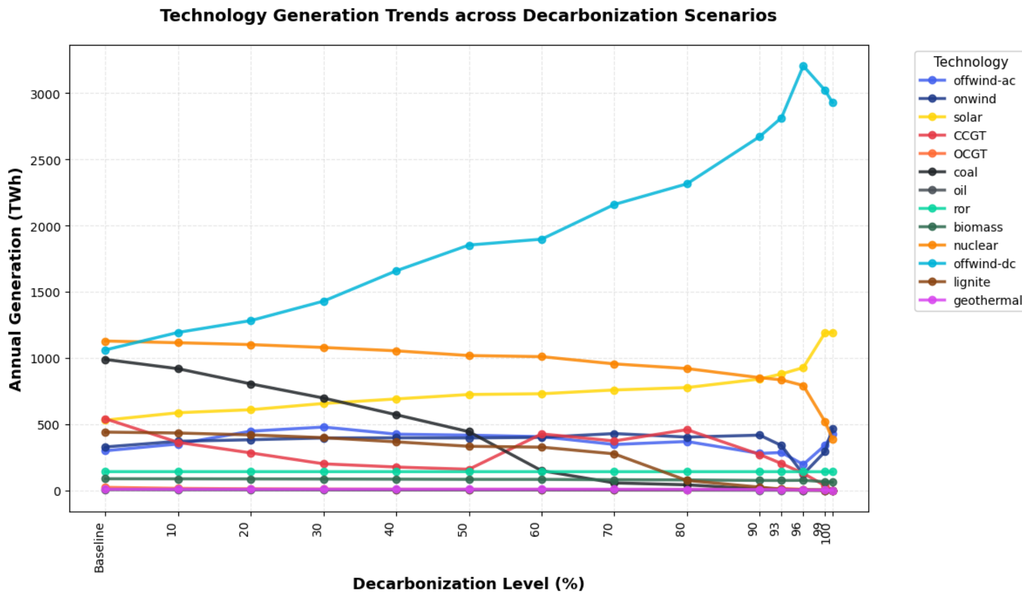

Figure 14: Decarbonized generator output by technology (line plot).

Here the generation dominance of off-shore wind at high levels of decarbonization is clearly visible. Interestingly at very high levels of decarbonization nuclear generation declines. this will be expanded upon when discussing the underlying mechanisms below. As for the transmission infrastructure in the decarbonized systems the toal transmission capacity seen below in figure 15:

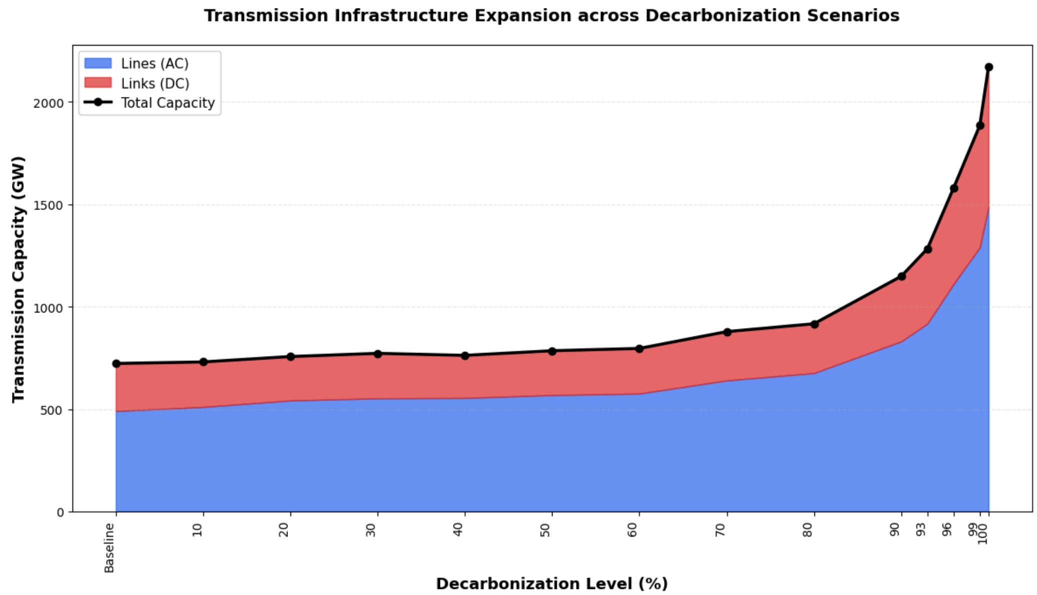

Figure 15: Decarbonized transmission capacity by technology.

The contribution to the total transmission capacity from the HVDC links remains fairly constant as the system decarbonizes, while the AC transmission capacity increases substantially. As expected the total transmission capacity sees a large increase at very high levels of decarbonization as the system relies more and more on non-dispatchable renewable generation which must be transported from the generation sites to the load centers or stored. The storage capacity evolution for the decarbonized systems can be seen below in figure 16:

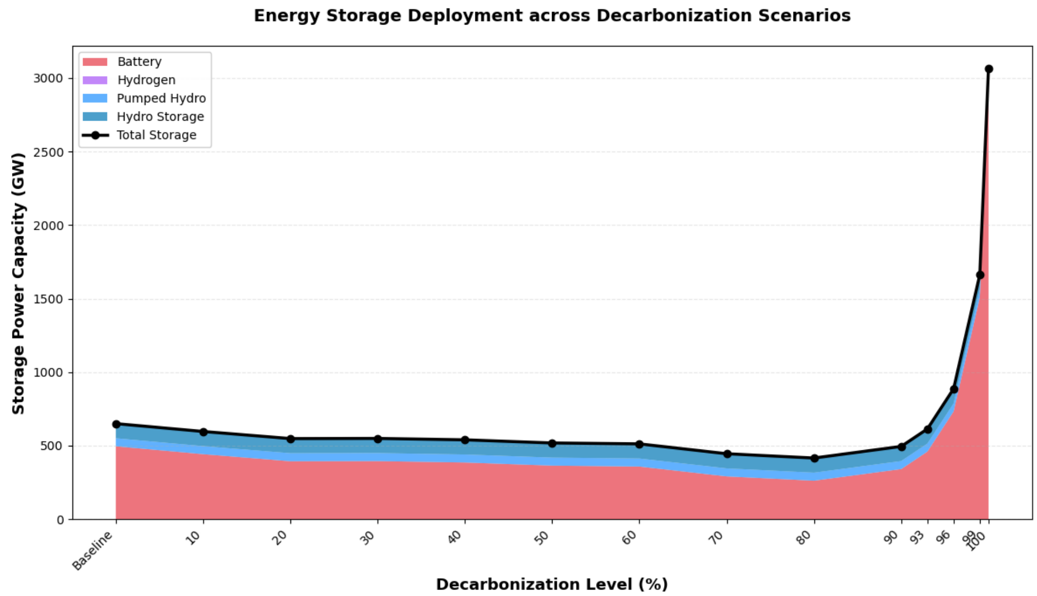

Figure 16: Decarbonized storage capacity by technology.

As expected a massive increase in storage capacity is seen at at the very highest levels of decarbonisation. This is caused by the decommisioning of the last dispatchable generation technologies which forces the system to use storage for nodal balancing where it has previously been able to rely on dispatchable fossil fuel based generation. Again the complete absence of hydrogen sstorage is notable, especially as the generation share of wind based generation is very high. This will be discussed further in the analysis section below. The emissions of the decarbonising systems are of course fractions of the baseline emissions as constrained by equation (15). The mean marginal demand-weighted price as defined by equation (22) can be seen in figure 17 below and the total system cost evolution for the decarbonized systems can be seen below in figure 18:

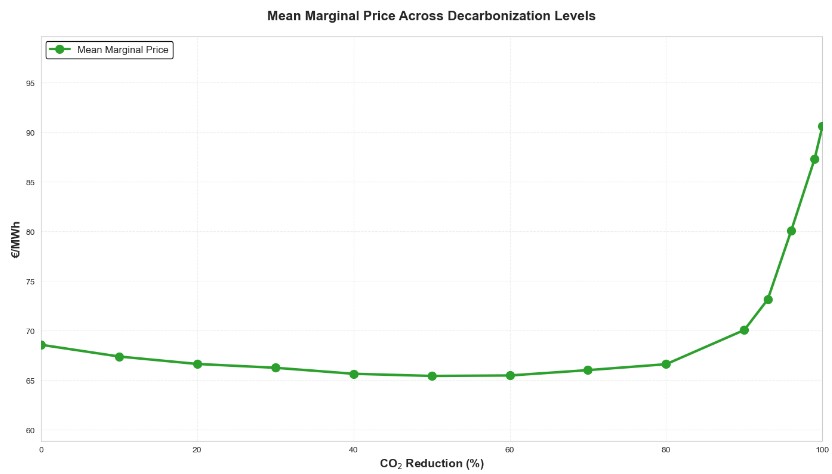

Figure 17: Decarbonized system economic metrics.

Interestingly the mean marginal demand-weighted price initially *decreases* as the system begins to decarbonize before increasing dramatically at high levels of decarbonization. While counterintuitive in a least cost optimized system, this is in fact not an artifact of the model but rather a result of the underlying mechanisms at play in the system. This will be discussed further in the analysis section below:

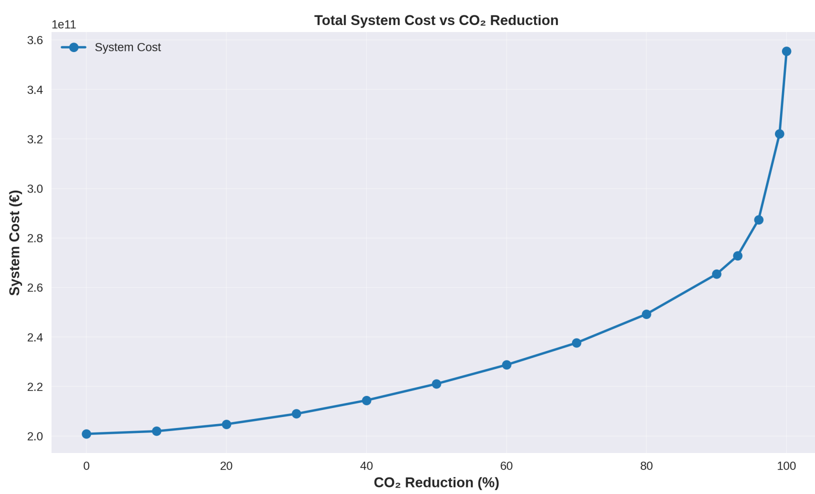

Figure 18: Total system cost in least-cost systems

The total system cost evolution as seen above behaves more intuitively. The total system cost increases quie dramatically from 200 B€ at the baseline to around 360 B€ at 100% decarbonization. The main contributor to this large increase is the large buildout of transmission- and storage-infrastructure as can be seen in figure 19 below.

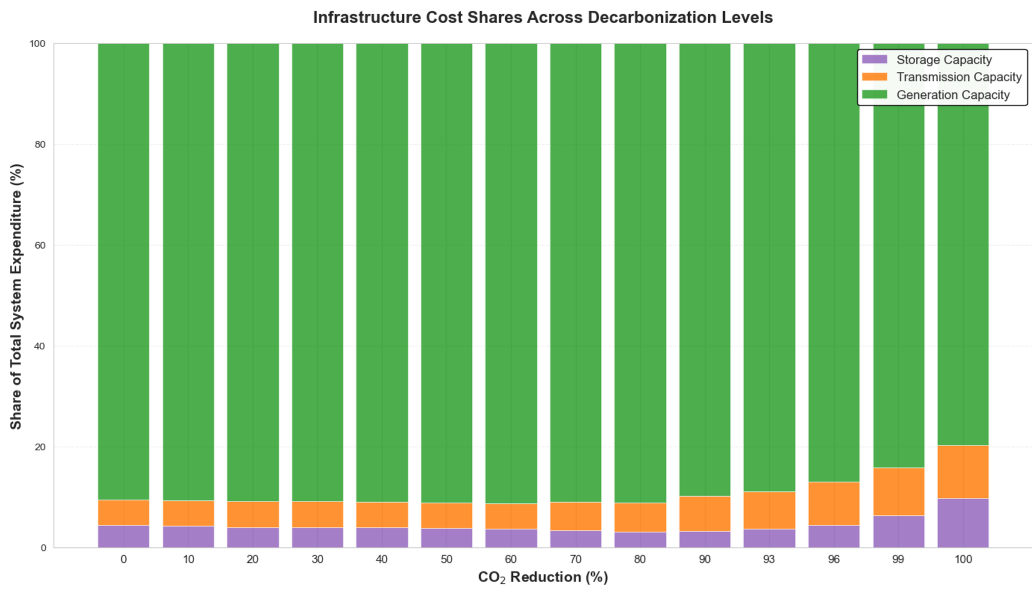

Figure 19: Decarbonized system cost share breakdown by category.

Finally the decentralization of the decarbonized systems can be quantified by the change in the equity measure (the SDRCL as defined in equation (21)) compared to that of the baseline. Seen below in figure 20 is the change of the equity for the decarbonized systems:

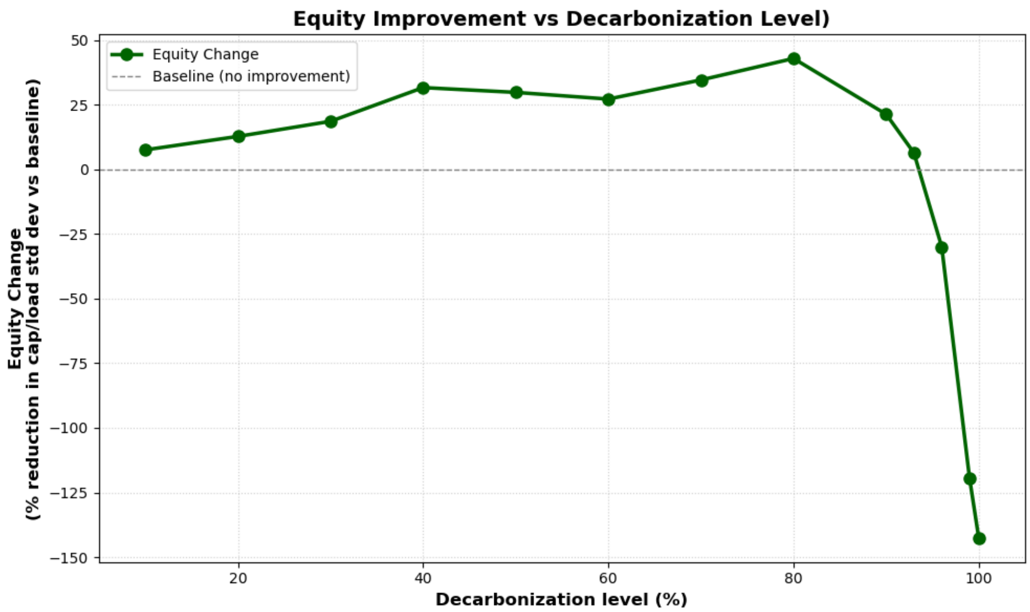

Figure 20: Decarbonized system equity change compared to baseline.

Here it can be seen that as the system decarbonizes the equity initially *increases*, meaning that the system becomes less centralized. However at high levels of decarbonization the equity totally collapses as the system becomes highly centralized. This will, of course be discussed further in the analysis section below.

4.4. Decarbonization mechanisms and interactions

The decarbonization process as seen above behaves mostly as one would expect. In general there seems to be a compunding effect in the system which means that as the system approaches 100% dearbonisation, changes between each iteration become more and more pronounced. A pronounced point of transition for many of the metrics shown above seems to be at 80-90% decarbonization where many of the metrics begin to change more rapidly. The generation mix of electricity be different technologies favors wind-based generation as these typially have a capacity factor time series (and ths generation) that lines up better with the load time series compared to the solar-based generation. This is caused by the fact that the load pattern in Europe tends to peak during winter when there are fewer hours of sunlight available to produce electricity. Transmission and storage infrastructure seems to be servicable up to around 90% decarbonization, after which a massive buildout is required in order to service the load with non-dispatchable renewable generation, which leads to a large increase in system cost. The system also overall becomes more centralized as the system decentralizes as expected, as the most optimal locations for renewable generation are utilized to a higher degree in order to meet the decarbonization constraints. However some few trends and results, as mentioned throughout the section above, do need further discussion

When reviewing the electricity prices associated with the optimized networks, it is of interest to compare two different cost measures: The mean marginal demand-weighted price and the levelized cost of electricity (LCOE). As a reminder the mean marginal demand-weighted price is defined as seen in equation (22) above. This represents the average price that consumers pay for electricity over the entire time period, weighted by their demand. The other cost measure, the levelized cost of electricity (LCOE) is defined as:

$$\text{LCOE} = \frac{\sum_{t \in T} \sum_{g} \left(\text{Capital Costs}_{g,t} + \text{Marginal Costs}_{g,t} \right)}{\sum_{t \in T} \sum_{n} L_{n,t}} \tag{23}$$

For generator technologies g. LCOE represents the average cost of producing electricity over the lifetime of the system.

Seen below in Figure 21 is a plot comparing the mean levelized cost of electricity for the different levels of decarbonization, compared to the mean marginal demand-weighted price:

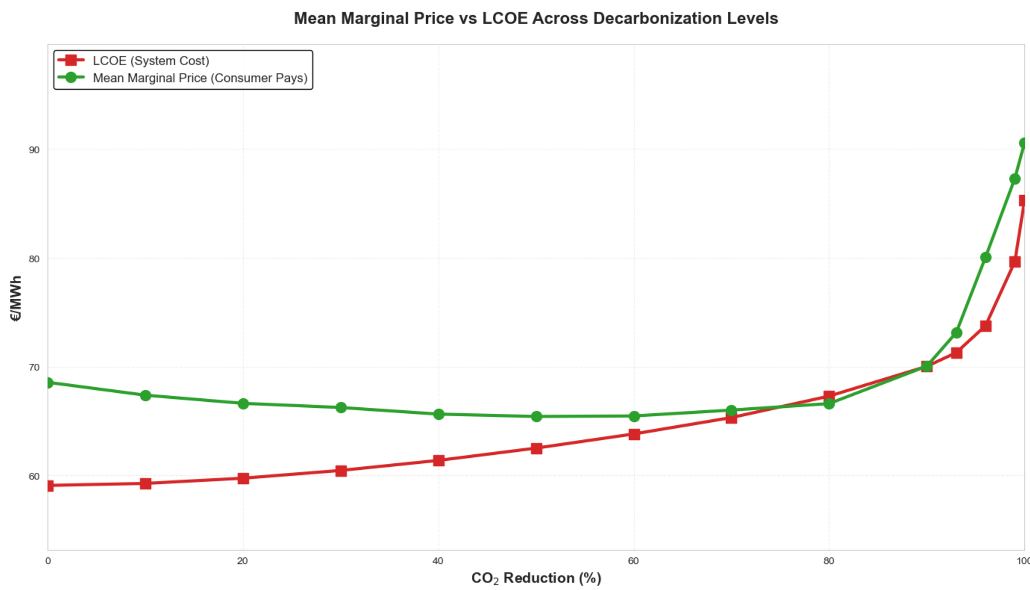

Figure 21: Comparison of mean LCOE and mean marginal price

This plot shows two different interesting results. Firstly, as commented on previously, it can be seen that as the system begins to decarbonize, initially the mean marginal demand weighted price *decreases* before increasing dramatically at high levels of decarbonization. Secondly it can be seen that for a majority of decarbonization levels a significant gap between the two cost measures is present.

Traditionally optimized energy system models such as the ones produced by PyPSA must adhere to the zero-profit condition [25]:

$$\sum_{t\in T}\sum_{n}\left(L_{n,t}\times\lambda_t\right)=\sum_{t\in T}\sum_{g}\left(\text{Capital Costs}_{g,t}+\text{Marginal Costs}_{g,t}\right) \quad (24)$$

This equation states that for all times the demand-weighted marginal price in all nodes must be equal to the sum of the capital and marginal costs across all technologies and times. Or, in other words: The revenue must be equal to the expenditure. However, critically, the zero-profit condition applies to long-run competitive equilibrium subject to the following assumptions:

- All capacity is free to expand and contract in order to achieve optimality.
- No artificially imposed constraints on the system.
- Steady state market conditions.

This clashes with the modelling methodology in several ways. The optimized model is a *Brownfield* system, meaning that even before optimization some fossil generation capacity exists in the system. As stated in the modelling section 3.5 , fossil generation capacity is allowed to contract but not to expand as constrained by:

$$p_{\text{nom}} \leq p_{\text{nom, max}} = p_{\text{nom, baseline}} \tag{25}$$

This means that the zero profit condition does not apply to optimization performed in this thesis, which leads to the difference between levelized cost and mean marginal price, also known as scarcity rent, as seen in Figure 21. The scarcity rents manifest when the optimizer ideally would increase capacity to meet demand, but is constrained by a binding capacity constraint.

Imagine for example that a fossil fuel-based generator is producing electricity at a lower marginal cost than the current market price. If the fossil based generator was allowed to expand, this would naturally attract additional fossil based generation to the market, until it becomes saturated and the mean marginal demand weighted price decreases to the point where the fossil generator is no longer profitable. However, as the fossil generator is not allowed to expand capacity, it can continue to produce electricity at a profit, leading to the scarcity rents seen in the system. This is the primary driving mechanism behind the gap between LCOE and mean marginal demand weighted prices seen in Figure 21.

This mechanism also applies to non-extendable renewable generation capacity, like nuclear, geothermal and biomass, as well as non-extendable storage like pumped hydro-storage. Pictured below, in Figure 22 is the evolution of the contribution to the total scarcity rent from the different technologies not constrained by the zero-profit condition:

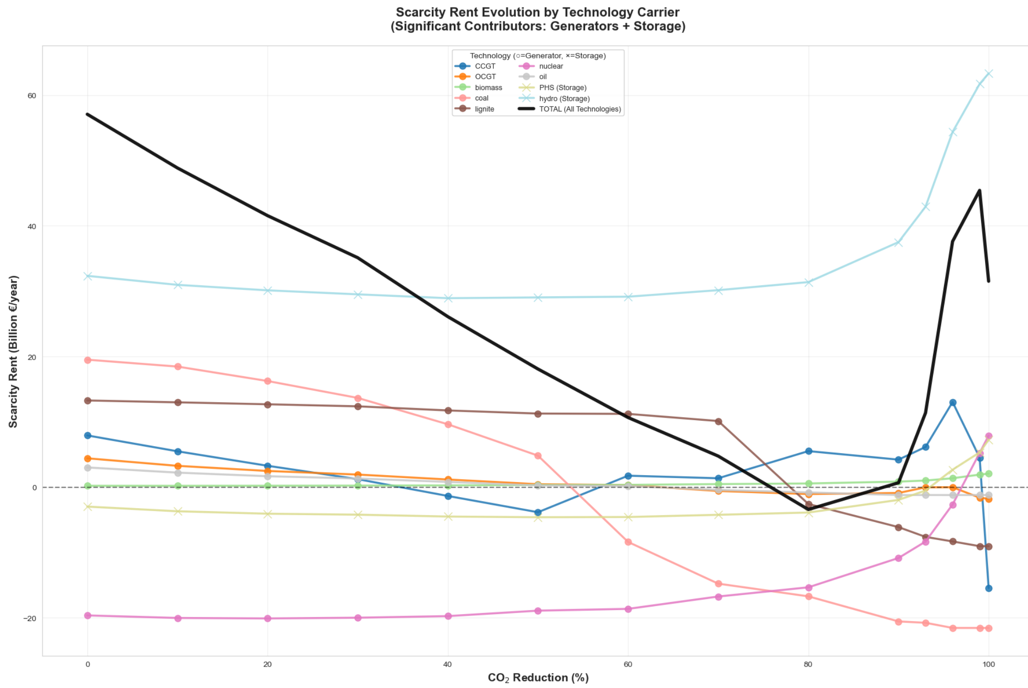

Figure 22: Scarcity rent contributions by technology

Here it can be seen that at low levels of decarbonization the majority of the contribution to the total scarcity rent comes from the fossil-fuel based generators. As the system decarbonizes the amount of fossil fuel generation is reduced in order to meet the decarbonization constraint and as such the contribution to the scarcity rent decreases. However at around 80% decarbonization where the vast majority of the fossil based generation has been phased out, the main contributors become the dispatchable renewable generators, such as nuclear, and storage units needed for balancing. This is illustrated in the modified version of Figure 21 below:

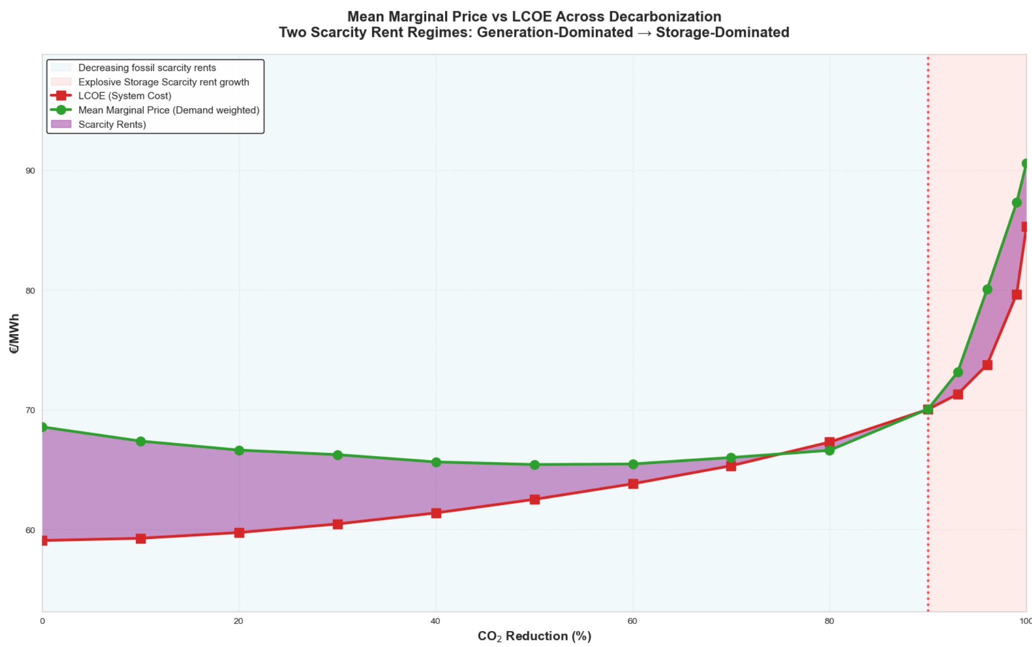

Figure 23: Transition between main scarcity rent contributors

As a side note, a negative scarcity rent indicates that the technology in question is running at a loss, meaning that the revenue from selling electricity is not sufficient to cover the capital and marginal costs of the technology. One might think that such a technology would not be included in the optimized system, however as the optimization is performed on a brownfield system with existing capacity, it may be optimal to keep such a technology running at a loss if the overall system cost is reduced by doing so.

As for the initial decrease in mean marginal demand weighted price. This can be attributed to something called the "merit-order effect": When the decarbonization constraint of fossil fuel energy generation tightens, fossil-fuel based energy production is replaced by renewable generation. As some forms of renewable generation have very low marginal costs the mean marginal demand weighted price decreases as these low marginal cost generators set the market price in more snapshots. However the total system cost increases as the newly installed generation- and storage-capacity has capital costs that contribute to the overall increase in system cost.

As for the centralization trends seen in figure 20 above, it is interesting that the equity inituially increases as the system bgins to decarbonize. The expectation would be that since renewable generation buildout is enforced through decarbonization measures these would be installed in the nodes with the bes renerable resources ad that the system would as a result of this see a continuos decrease in equity. In order to uncover the underlying mechanism that leads to the inital decentralization when decarbonising the system, a spatial correlation is defined as seen in equation (26) below:

$$\rho = \frac{\sum_n (C_n^R - \langle C^R \rangle) \cdot (L_n - \langle L \rangle)}{\sqrt{\sum_n (C_n^R - \langle C^R \rangle)^2} \cdot \sqrt{\sum_n (L_n - \langle L \rangle)^2}} \tag{26}$$

Here C_n^R is the installed renewable capacity at node n, L_n is the load at node n and $\langle \cdot \rangle$ denotes the mean value across all nodes. This correlation is thus an indicator of how well the installed renewable capacity aligns with the load distribution in the network. A higher positive correlation indicates that the renewable capacity is more evenly distributed across the nodes relative to their load, while a negative correlation indicates a more uneven distribution. Seen below in figure 24 is the evolution of the spatial correlation as the system decarbonizes:

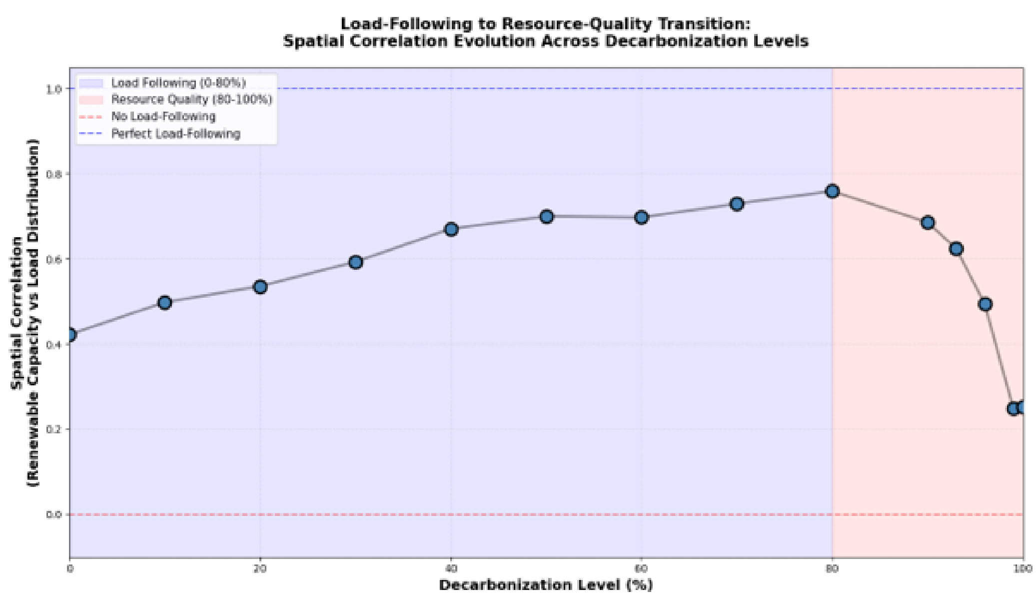

Figure 24: Spatial correlation between renewable capacity and load as the system decarbonizes

As the system decarboizes the spatial correlation between the load and the installed renewable capacity increases up untill 80% decarbonization where it suddenly decreases to below the orginal level. This seems to indicate that there are two separate mechanisms or paradigmes at play here. At lower levels of decarbonization the new installed renewable generation capacity is foollowing the load centers in the network. When compared to the baseline system this leads to more capacity being installed in more diverse location and therefore increasing equity. However at very high levels of decarbonization where renewables dominate the system and backup from dispatchable based generation decreases, the system begins to increasingly optimize for the best renewable resources in order to minimize costs. This centralization of renewable capacity can also be seen directly when observing the capacity maps for the two extremes at 80% and 100% decarbonization seen below in figures 25 and 26 respectively:

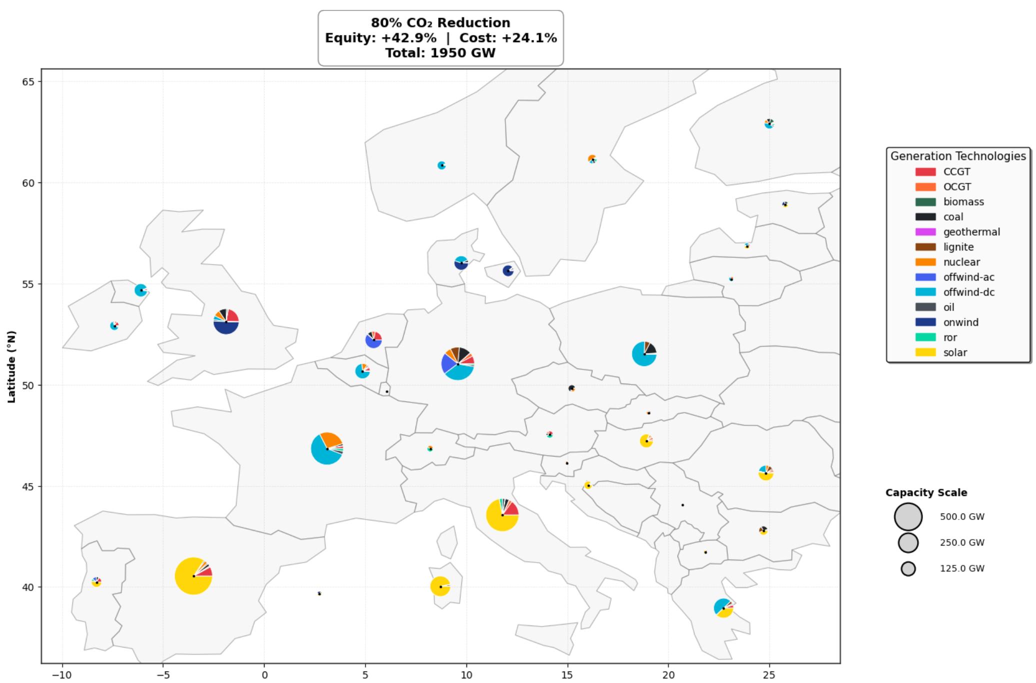

Figure 25: Generation capacity map at 80% decarbonization

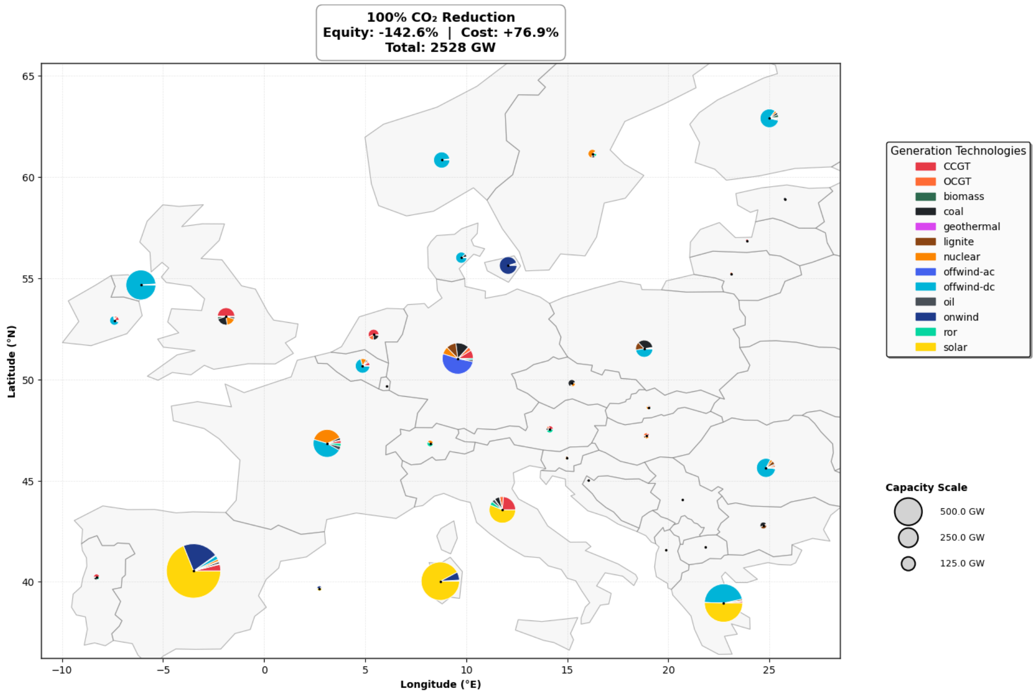

Figure 26: Generation capacity map at 100% decarbonization

When observing regions with good renewable resources such as offshore wind in Northern Ireland and solar in Spain, it can be seen how at 100% decarbonization the favored renewables have been utilized to a much higher degree than at 80%. On the flipside regions with less favorable renewable resources have seen their installed renewable capacity decrease like in the northern Balkan region. Should i discuss centralization of storage as well? maybe just a few sentences stating that storage follows generation?

As can be seen in figure 15 above as the total transmission capacity increases, this buildout mainly consists of the AC lines while the HVDC links remain fairly constant at the different levels of decarbonisation. This is due to the fact that HVDC links are mainly used for long-distance or cross-ocean transmission. while the AC lines are mainly used in mainland Europe as a part of the main contenental mesh Network. As the system decarbonizaes the optimizer prioritizes the expansion of the AC part of the network as this is what benifits more nodes in the system. This then in turn provides more flexibility in the location of storage and generation compared to expanding the HVDC network. However at the very extreme of 100% decarbonization the HVDC capacity does see a small increase as the system now needs to use the renewables in the best resource locations regardless if this in mainland Europe or, for example, in the UK where offshore wind resources are very good.

In figure 14 above there seems to be a trade-off at around 90% decarbonization where the system suddenly priorities DC-offshore wind instead of AC-offshore generation. Then at even higher levels of decarbonisation this trade-off then gradually disappears, and the levels at 100% decarb end up being fairly consisten with the rest of the trend-line. This seems to be an artifact of the discrete steps taken in the modelling of the decarboinization results. As the CO_2 constraint tightens the viable solution space shrinks which can lead to sudden shifts in the optimal solution as seen here. On the other hand the sudden decline in the amount of nuclear generation at high levels of decarbonization is not an artifact but an interesting mechanism. The sudden decline in nuclear generation coincides with the large buildout of the storage capacity and seems to indicate that the syztem transitions from being dominated by one paradigme to being ominated by another paradigm. Nuclear generation is dispatchable and such has primarily acted as a way of balancing the system when the non-dispatchable renewables are not producing enough generation. As the system is forced to build storage by some other mechanism at high levels of decarbonisation lkess nulear generation is needed for balancing, which leads to the decline.

The absence of hydrogen based storage at all levels of decarbonization is not intuitive as one typically associates hydrogen storage with long-term storage and wind based generation which is more seasonal in nature. However it seems that the poor round-trip effeciency of the hydrogen storage means that it is not cost competetive with battery storage even at high levels of wind based generation. The rate o which wind based generation is build out trends towards wind-based generation being prioritized increasingly at higher levels of decarbonization. This also coincides with the sharp increase in scarcity rent of hydro storage as seen in figure 22 above. This can be interpreted as he fact that, if unconstrained, the optimizer would have increase water-based storage capacity which is a form of long-term storage and as such is better suited to balance wind based generation.

4.5. Decentralized systems

Having performed pertubations of both the level of decarboonization and the decentralization constraint K yielding a total of 105 networks means that in order to summarize the results some initial screening is required. For the capacity results this has been done by performing a coefficient of variation (CV) analysis across the different K-constraints for each decarbonization level. the results of which can be found in appendix Appendix C: Capacity sensitivity. Unsurprisingly the generation capacity analysis shows that the freely extendable renewable generation technologies solar and wind are impacted the most. Pictured in figure 27 below the evolution of the total installed renewable generation capacity for the different levels o decentralization and decarbonization. Interstingly, it can be observed that the extreme scenario of complete decentrralization (K=1) is not consistant with the trends of decentralization. The color of the background indicates if the capacity for the fully decentralized system (K=1) is consistant with the general trend when decentralizing or not.

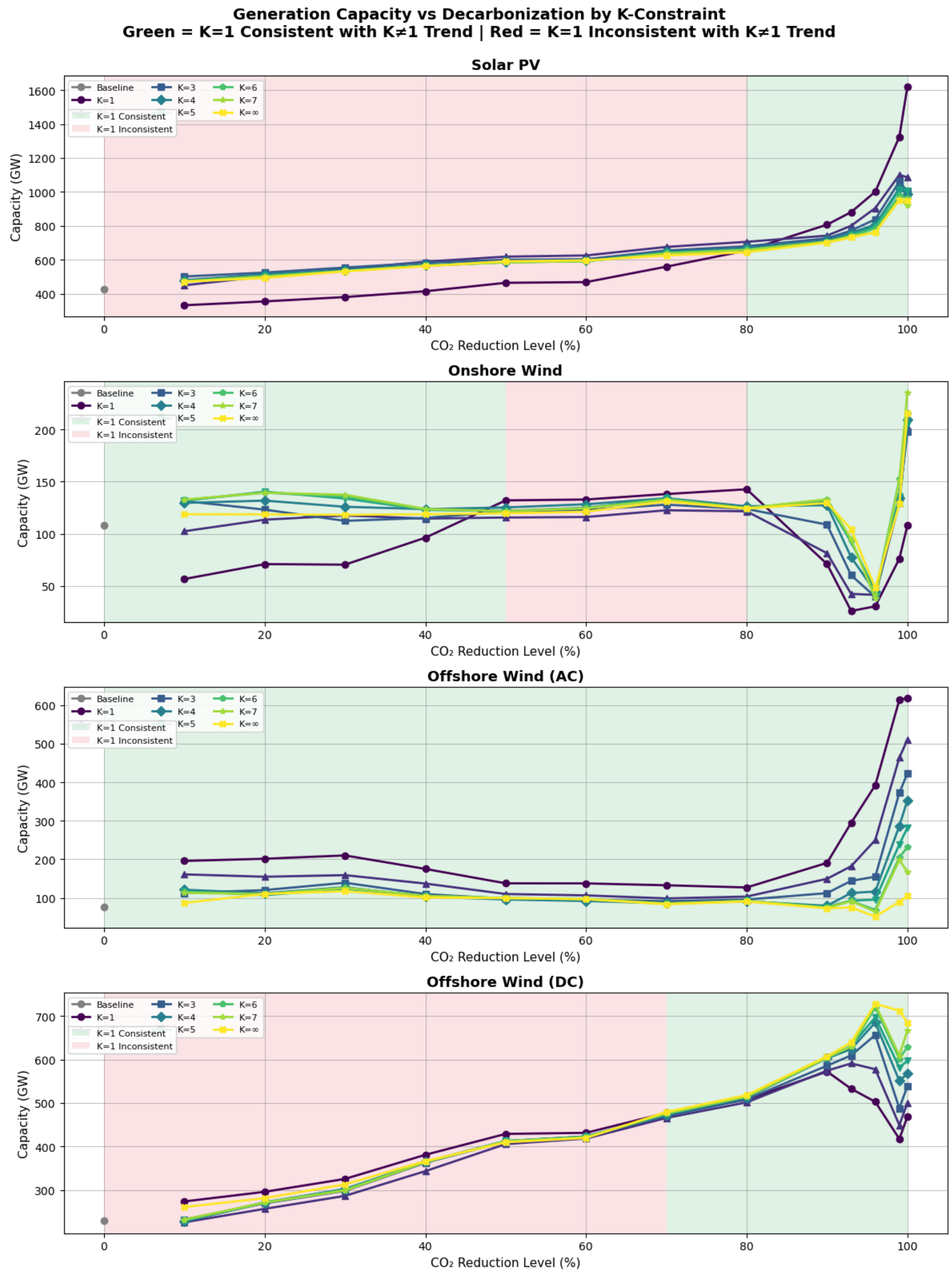

Figure 27: Total installed renewable generation capacity across K-constraints and decarbonization levels.

Here it can be observed that The K=1 systems is highly volatile when compared to the other levels of decarbonization. This may indicate that the highly resstrictive nature of the K=1 constraint leads to rapid changes in optimal decarbonisation straegies depending on the level of decarbonisation. The main observations can be seen below:

- **Solar PV:** When decentralizing the system, in general a more decentralized system leads to an increase in solar capacity, regardless of decarbonization level. However the K=1 scenario diverges from this trend at decarbonization levels below 80% while at higher levels of decarbonization it becomes consistent with the trend.
- **Onshore Wind:** When increasing decentralization, installed capacity decreases. This effect is particularly pronounced at high levels of decarbonization. At low levels of decarbonization the K=1 is consistant but amplified before diverging from 50% to 80% before finally becoming consistant again at higher levels of decarbonization.
- **Offshore Wind (AC):** A more decentralized system leads to more installed capacity across all levels of decarbonization with a significant increase at low levels of decarbonizatio amd a huge increase at high levekls of decarbonization. The extreme K=1 scenario is consistent with this trend at all levels of decarbonization.
- **Offshore Wind (DC):** In general a decentralizing system will lead to an decrease in installed capacity, particularly at high levels of decarbonization. At low levels of decarbonization the K=1 scenario is inconsistent with this trend but at high levels of decarbonization it becomes consistent.

5. Sanity checks and validations

Comparing the key figures like marginal prices to those from real networks

6. Discussion

Introduction to discussion

6.1. Economic trends and societal paradigms

What can interactions can be taken from my results and how does this align with expectations and real-world trends

6.2. Future policy implications

Discussing how this workflow can be used to analyze and incorporate paradigms to review implications of energy policy.

6.3. Uses for methodology

How else could this approach be used and how would this expand the workflows use-case.

7. Conclusion

8. Assumptions

9. Acknowledgments

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11. Appendices

11.1. Appendix A: Generator cost assumptions

11.2. Appendix B: Storage cost assumptions

This complicates scarcity rents as hydro does not have a captal cost defined

Table 6: Generator technology cost assumptions used in the optimization model.

Technology	**Capital Cost (EUR/MW/a)**	**Marginal Cost (EUR/MWh)**
CCGT	99.0k	47.20
OCGT	45.1k	58.38
biomass	379.2k	14.96
coal	180.2k	24.10
geothermal	490.7k	0.00
lignite	180.2k	13.49
nuclear	535.5k	16.90
offwind-ac	196.7k	0.01
offwind-dc	219.4k	0.01
oil	37.9k	130.23
onwind	133.8k	0.01
ror	270.9k	0.00
solar	55.1k	0.01

Table 7: Storage technology cost assumptions extracted from the optimized network.

Technology	**Capital Cost (EUR/MW/a)**	**Marginal Cost (EUR/MWh)**
Battery	13.7k	0.00
Hydrogen	205.0k	0.00
PHS	160.6k	0.00
Hydro	0.0k	0.00

Table 8: Installed generation capacity across decarbonization levels. Values show mean capacity (GW) with coefficient of variation (%) in parentheses. Cell shading indicates K-sensitivity: green = capacity increases with K (favored by decentralization), red = capacity decreases with K (disfavored by decentralization), unshaded = insensitive to K-constraint. Darker shading indicates stronger sensitivity.

Technology	0%	50%	80%	100%
Solar PV	424.3 (0%)	564.7 (11%)	666.2 (4%)	1166.1 (23%)
Offshore Wind (DC)	229.8 (0%)	414.2 (2%)	510.4 (1%)	562.5 (15%)
Offshore Wind (AC)	76.1 (0%)	111.5 (14%)	103.3 (14%)	378.8 (53%)
Onshore Wind	108.5 (0%)	122.3 (5%)	128.0 (7%)	182.7 (24%)
CCGT (Gas)	156.0 (0%)	156.0 (0%)	156.0 (0%)	156.0 (0%)
Nuclear	130.4 (0%)	130.4 (0%)	130.4 (0%)	130.4 (0%)
Coal	119.6 (0%)	119.6 (0%)	119.6 (0%)	119.6 (0%)
Lignite	50.2 (0%)	50.2 (0%)	50.2 (0%)	50.2 (0%)
OCGT (Gas)	39.7 (0%)	39.7 (0%)	39.7 (0%)	39.7 (0%)
Run-of-River Hydro	34.5 (0%)	34.5 (0%)	34.5 (0%)	34.5 (0%)
Oil	32.0 (0%)	32.0 (0%)	32.0 (0%)	32.0 (0%)
Biomass	9.9 (0%)	9.9 (0%)	9.9 (0%)	9.9 (0%)